$$
\begin{aligned}
\ell_j = & -\tau \int \phi_j''' X \, dt + (1+\kappa\tau) \int \phi_j'' X \, dt \\
& -(\kappa+\gamma\tau) \int \phi_j' X \, dt + \gamma \int \phi_j X \, dt, \\
u_j = & \int \phi_j e \, dt, \qquad \widehat{c} = \max\left(0, \frac{\sum_j u_j \ell_j}{\sum_j u_j^2}\right), \\
D^{\text{demo}}_{\text{EEG}-\text{Hb}} = & \frac{\sum_j (\ell_j - \widehat{c} u_j)^2}{\gamma^2 \sum_j (\int \phi_j X \, dt)^2}.
\end{aligned}
$$